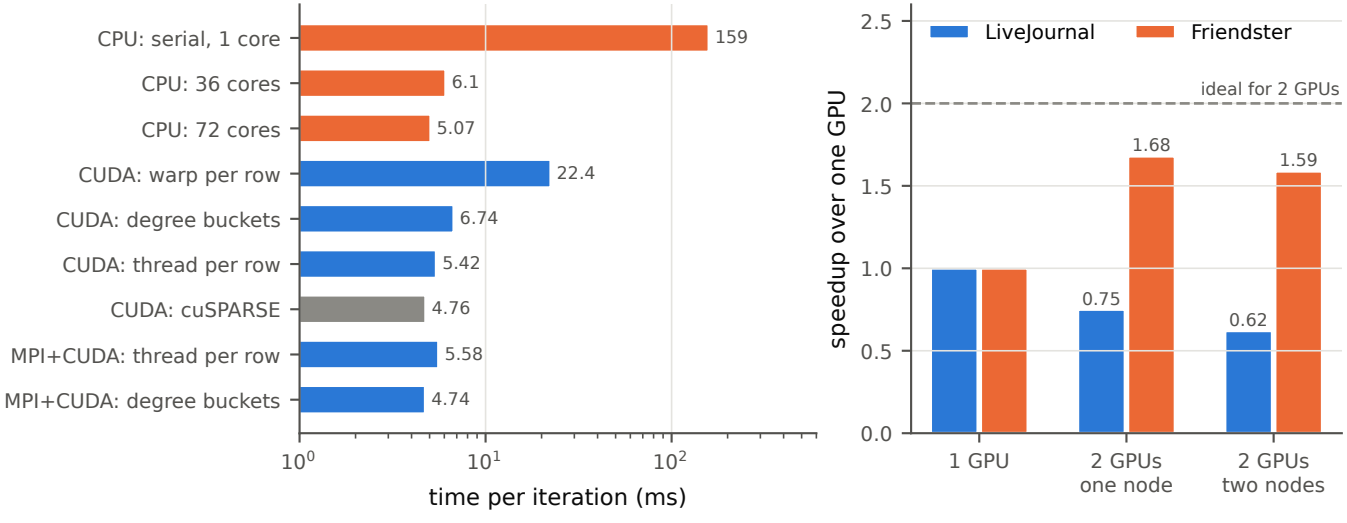
(a) soc-LiveJournal1: CPU and one A40
CPU: serial, 1 core
CPU: 36 cores
CPU: 72 cores
CUDA: warp per row
CUDA: degree buckets
CUDA: thread per row
CUDA: cuSPARSE
MPI+CUDA: thread per row
MPI+CUDA: degree buckets
159
6.1
5.07
22.4
6.74
5.42
4.76
5.58
4.74
10^0
10^1
10^2
time per iteration (ms)
(b) MPI+CUDA on two GPUs
LiveJournal
Friendster
ideal for 2 GPUs
speedup over one GPU
2.5
2.0
1.5
1.0
0.5
0.0
1.68
1.59
0.75
0.62
1 GPU
2 GPUs
one node
2 GPUs
two nodes